Problem Statement

The retail industry is undergoing a significant transformation as consumers increasingly shift between online and physical store shopping. Businesses struggle to understand *why* a customer chooses one channel over another — is it their age, income, tech-savviness, or sensitivity to delivery fees? Without a data-driven model to predict individual shopping preferences, retailers must rely on broad demographic assumptions, leading to poorly targeted marketing campaigns, suboptimal inventory allocation, and missed revenue opportunities. The challenge is to build an intelligent system that can accurately classify a customer's shopping preference — **Online**, **In-Store**, or **Hybrid** — using their behavioural, demographic, and attitudinal attributes.

To address this, a machine learning classification model must be trained on the *Online vs Store Shopping Dataset* (~11,789 customer records, 24 features) covering variables such as monthly income, daily internet hours, tech-savvy score, online payment trust, need-to-touch-feel score, and city tier. The model must automatically preprocess mixed feature types, compare multiple algorithms (Logistic Regression, Random Forest, Gradient Boosting) via cross-validation, select the best performer, and expose its predictions through an interactive Streamlit web application — enabling any business user to enter a customer profile and instantly receive a shopping preference prediction along with class-wise probability scores, without any programming knowledge required.